Applying Formal Software Engineering Techniques to Smart Grids

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***Abstract*—Engineering complex systems that have to meet critical requirements is a difficult task, especially if multiple engineering disciplines are involved. Common practice in domains like the automotive or avionic industry shows that formal methods improve engineering process efficiency for embedded software due to abilities like abstraction, early verification and iterative refinement. This paper presents how existing formal software engineering methods can be adapted to meet the needs of the smart grid domain. A case study demonstrates how we develop a basic interdisciplinary but semantically integrated decomposition of a household including electric and software behavior. Finally, we provide first simulation results to evaluate the feasibility of the model as well as the presented engineering method.**

***Keywords*-Systems Engineering; Smart Grid; Smart Energy Systems; Model-based Development; Formal Methods; System Architecture; Simulation;**

I. Introduction

The increasing amount of energy from renewable sources forces existing infrastructures to deal with more fluctuating supply. To meet this challenge, smart grids integrate information technology into existing electrical energy systems [1]. They provide bidirectional interaction between suppliers, consumers and storages in order to sustain or improve the environmental efficiency, security of supply, and economic efficiency of the energy system. As smart grids will reach large scales, are highly distributed, behave in very dynamic ways and have to meet several critical requirements, it is a major goal to reduce the overall complexity inherent in these systems and thus to increase its feasibility and transparency. To ensure a clear understanding, systems have to be modeled in accurate ways. Formal engineering methods offer the possibility to form semantically meaningful, consistent models of systems. Those can be used to analyze system wide behavior even in early stages of the development process. Starting with abstract high level models, the system is gradually refined by systematically adding details to models. Decompositional theories in conjunction with the use of verification methods allow to derive behavioral properties of composition. This is beneficial to detect problems existing in the system design and can be used to verify systems even in large scales. The starting point for this challenge lies in the current energy system, which consists of a physical infrastructure and several legacy systems. This already existing system has to be step-wise transformed, supplemented in order to meet the new requirements. Accordingly, new systems will be introduced, old systems will be adapted or used in another way. To model an overall system behavior, a reverse engineering of the old system and its new additions is needed. This has to be taken to account when trying to form a general modeling theory.

Problem Statement: The development of systems belonging to the smart grid is a highly interdisciplinary challenge, including various requirements from different domains like power systems, networks & communication, software engineering, business and human interaction. All these disciplines will influence the behavior and structure of the smart grid on different stages and with different impacts. Also other impacts not existing in traditional information systems, like weather or power nets, will have a direct impact on systems belonging to the smart grid. To understand the overall system behavior all those influences have to be taken into account and modeled explicitly.

Research Objective: Our goal is to answer the question how existing methods can be adapted to model different aspects relevant for the smart grid. To which degree is it necessary to adapt existing formal methods? What kind of abstractions have to be made in order to model a specific behavioral aspect existing in the smart grid? Finally, what kind of structure can be used to describe smart grid systems? We think that these aspects are useful to understand the future work on formal methods, which can be used to model smart grids.

Contribution: In this paper we present a first experience report on how to adapt existing formal, component-based methods focusing on households and their involvement in low voltage nets as a case study, which we classify as a relevant part of the smart grid. We show how to build a formal system model including physical system, control system and cross-cutting concerns. Additionally, we present a basic physical system model which is able to express digital as well as physical properties. Finally, we use our model to simulate the behavior of a household and discuss how our simulation results can be interpreted.

II. Related Work

Over the past decade we have witnessed a number of research efforts to understand and develop large scale systems through component-based engineering methods [2],

[3]. The main reason for applying such techniques is to reduce the systems' structural complexity while increasing reuse, therefore making systems more manageable. However, due to ever increasing system scales existing techniques are reaching their limits of feasibility. One reason is that verfying properties of such large-scale systems is still a challenging due to exploding state spaces and the computational complexity invloved. A second reason is that aspects from different engineering disciplines such as mechanical, electrical and software engineering have to be formalized in a common framework to model and analyze the overall system behavior. Literature provides a variety of component-based engineering methods, which are able to capture and integrate aspects from multiple engineering disciplines. In the following prominent examples are discussed with respect to the aim of this paper.

Aligned with object-oriented software engineering Göhner et al. [2] propose a component-based engineering approach for automation systems. The proposed method decomposes embedded systems into components, uses object-oriented notations and implements the software fraction in conventional programming languages. Consequently, Eberle et al. [3] find that such component-based engineering methods effectively reduce complexity and increase manageability. Nevertheless, they state that difficulties in adapting component-based approaches to embedded systems with hard real-time requirements exist. Also, the existing component-based is not yet extended to a formal interdisciplinary approach.

A respective trend targeting system development and system evolution in automation engineering is the concept of mechatronic modules as for example introduced by Bonfe et al. [4]. Consequently, the concept of mechatronic modules forms the basis of a true multi-disciplinary component-based engineering method, where automation systems are composed of predefined building blocks. The predefined building blocks integrate engineering artifacts from different disciplines such as mechanical engineering and electrical engineering with complementary software components. However, at the current stage formal model semantics are limited to the respective application domain and a generalization is non-triviral.

The de-facto industry standard for modeling systems is the Unified Modeling Language (UML) [5]. The main disadvantage of the UML is the lack of formal semantics, which is why the possibilities for automatically analyzing UML system models with tools like simulation, model checking or theorem proving are limited. An exception are UML state machine diagrams, a derivative of the original state charts as introduced by David Harel [6] for which precise behavioral semantics can be defined [7], [8]. Still, overall UML does not provide a strong formal system-theoretic foundation, which in practice mostly results in underspecified and inconsistent system specifications.

An extension to the UML specifically for systems engineering is the SysML [9]. Shah et al. [10] propose a multi-view modeling approach based on SysML for requirements engineering in early phases along with SysML domain specific profiles for discipline related modeling. They show that SysML can be exploited for high-level requirements specification, modeling of system structure and behavior as well as their dependencies enabling traceability of decisions. Further, an approach for checking consistency between models of different disciplines is provided.

Another approach for multi-view modeling is the so-called Model Integrated Mechatronics (MIM) [11]. MIM provides an architecture with several views on a mechatronic system with a mechatronic view on the top level. The base component of MIM is the mechatronic component called MTC. Nevertheless, MIM strongly focuses on the software view. Based on MIM Thramboulidis [12] introduces modeling of mechatronic systems in a 3+1 layer model based on SysML. This MTS model is surrounded by four views representing all disciplines participating in the engineering process.

Despite the advantages of SysML-based system engineering the problem remains that the UML profiles cannot replace semantic aspects of the UML. Therefore, SysML models are based on the same formally underspecified UML semantics.

III. Engineering Method

We base our work on FOCUS [13], a system engineering method featuring formal specification of structure and behavior as well as step-wise decomposition and refinement. The core of the method is formed by a layered reference architecture for embedded systems including functional, logical and technical aspects [14]. The individual layers provide independent but integrated viewpoints onto the system and guide the development process from requirements specification to deployment.

To adapt the method to the energy domain in this work we investigate the feasibility of the underlying system modeling theory for describing the structure and behavior of smart grids including both physical and digital aspects. The basic building blocks for describing the structure of systems are components and channels. Components encapsulate system behavior and use typed asynchronous directed channels to communicate with their environment. The concept of channel histories is defined to represent the channel valuations over time (also called traces). Consequently, component behavior is defined as a mapping from input to output channel histories.

Methodically, channels are typically used to carry digital information such as sensor measurements or messages exchanged via a communication medium. To model the structure and behavior of non-digital system components in interdisciplinary settings like the smart grid domain we therefore introduce the following distinction: (a) **Information channels** are used at the interface of digital components such as software and carry digital messages. (b) **Observation channels** are used at the interface of non-digital components such as physical systems or users and carry non-digital messages relevant for the system behavior such as physical properties or interaction events. It should be noted that observation channels are also modeled as typed channels, i.e. regular data types are used to describe the observations.

To specify encapsulated component behavior two choices exist: (1) Components can be decomposed into a set of subcomponents and internal channels, which are responsible for implementing the I/O mapping. Or (2) component behavior can be specified directly through formalisms like timed automata or temporal logic. In the case of decomposition the component behavior can be derived formally from the behavior of the respective subcomponents. Consequently, verification of component behavior can be derived from properties of sub-component behavior leading to a divide-and-conquer verification strategy [15].

One of the most common formalism for directly specifying component behavior are timed automata. Timed automata are defined in terms of states, transitions, variables and actions. Variables can be used to store values internally for calculations across several time steps. Actions assign valuations to output channels in each time step depending on input channel valuations and variables. Both weakly causal and strongly causal semantics are supported. States define actions, which are executed while in the respective state. And transitions define possible state changes depending on guard expressions. Note that more than one transition might be enabled at a time leading to non-deterministic models. Finally, an initial state is specified where generation of component behavior depending on input channel histories starts.

IV. SYSTEM MODEL

The following case study shows first results on using FOCUS formalism to describe the logical structure and behavior of energy systems. Physical legalities are reduced to the characteristic of produced and consumed energy within a given time frame. Effects like reactive energy and network constrains are neglected. However, this simplification has two positive effects: (1) The model complexity with respect to its parameters is decreased and (2) the model scalability with respect to (automated) verification and validation is increased. Also, today's energy markets are based on similar abstractions, which speaks for a valid abstraction [16].

A. *System Architecture*

The general idea for applying formal engineering to the energy domain is to encapsulate physical components with a formal component interface as depicted in Figure 1.

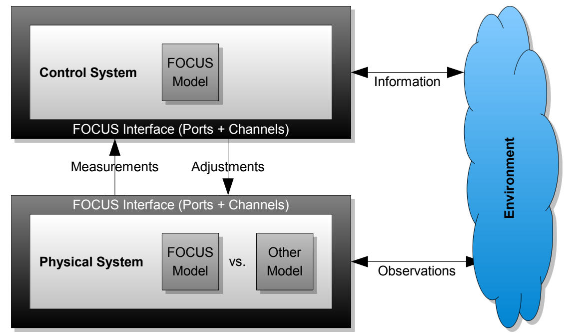


Figure 1. Physical system model encapsulated by FOCUS interface for information and observation exchange with its environment.

Consequently, the behavior of such components is described in terms of input and output information/observations over time, which enables integration with control logic via respective communication channels. In particular, the channels are used to communicate sensor measurements such as temperature and actuator adjustments such as device activation. Furthermore, other observations such as environment parameters or user interaction events can take direct influence on the behavior of the physical system.

Internally, the structure of the physical system (i.e. electric net) is simplified to resemble a tree as shown in Figure 2. The nodes of the tree represent the individual net sections from maximum-voltage to household and device level. Further, it is assumed that along the edges physical properties such as voltage or power can be measured. Consequently, Section IV-B describes a basic model of the measurements over time using the FOCUS formalism. However, other models such as differential equations could be used instead. In general, the selected model depends on the porpose of system modeling. E.g. for large-scale system simulation basic physical models with few parameters might be preferred,

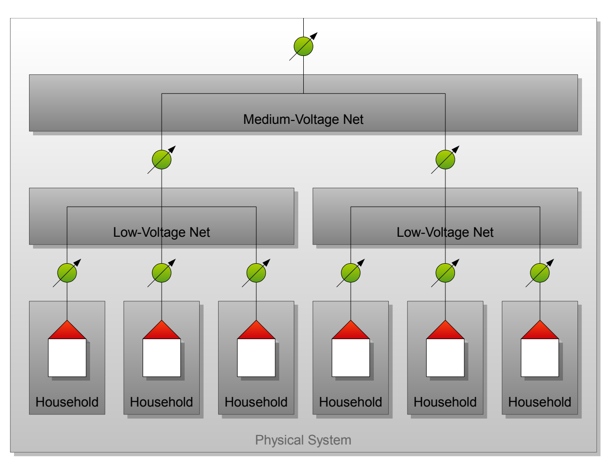


Figure 2. Tree representation of electric net topology with measurement points along the edges.

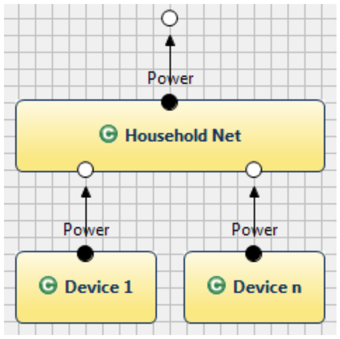


Figure 3. Decomposition of the physical system into components connected by observed power in discrete time intervals.

whereas detailed models could be used for analyzing isolated component behavior.

B. Physical System

The representation of the physical system (see Figure 3) serves as the foundation for developing the control logic. For simplicity this case study is focusing on a single household. Power is observed for discrete time intervals at various points in the electric net. In particular, each device produces/consumes power while the household net aggregates the respective inputs to a single output power. Power is represented as a number with the sign indicating the energy flow direction: The positive sign is used when energy is produced, the negative sign is used when energy is consumed instead. It should be noted that the energy flow direction is independent of the channel direction. The channel source is responsible for assigning the observed power, whereas the channel target behavior depends on the observed power.

The behavioral specification of the household net can be achieved with an automaton containing a single state `aggregate`. When in this state the observed power ouput at timepoint t is equal to the sum of the respective input power observations, i.e.

$$P_{out}(t) = \sum_{1 \leq i \leq n} P_{in}^{i}(t) \tag{1}$$

Consequently, at the maximum-voltage net this sum is an indicator for the overall net balance. I.e. a positive number is evidence for too high production, a negative number for too high consumption instead.

The behavioral specification of devices depends on the actual device types. For example, a basic lamp can be specified easly with a pair of `on`/`off` states. In `off` state the lamp defines an observable power of $0W$ while in `on` state the lamp defines an observable power of $-100W$. Transitions allow to switch between these two states. Typically, the transitions are triggered by the user.

In contrast, the behavioral specification of a refrigerator is much more involved. Typically, a cooling level and a sensor measuring the interior temperature are involved. For future smart grids the behavior probably will depend on some form of software controller including market and user preference data. This case is described in Section IV-E.

In summary, the overall model yields a basic abstraction of the underlying observable physical truth neglecting physical properties such as voltage or frequency. The computational complexity for each net component is linear in the number of inputs and therefore highly scalable. The computational complexity for each device is typically constant due to assuming constant electrical resistance and constant voltage.

C. Control System

The next step is to enrich the purely physical system model with digital (i.e. smart) control logic. One question that arises is where to place digital information channels and respective behavioral specifications. Obviously, real-world products such as refrigerators already contain digital components. Therefore, integrating physical and digital parts into a single compound seems the natural choice. However, a disadvantage of this strategy is that the structure of the control system has to follow the structure of the physical system. This is not always the case, e.g. when a service provider aggregates households from different net parts. Hence, for the case study the control system and the physical system are separated into independent structures. The channels are intended to transport sensor measurements such as temperature and actuator adjustments such as cooler activation/deactivation.

In general, control system modeling does not defer much from traditional formal software system modeling, in particular for the embedded domain, which is why more detailed models are omitted here. However, it should be noted that decomposition strategies have to be chosen carefully for the energy domain to yield proper solutions with desired quality characteristics.

D. Environment

Similar to the physical and the control system there are a number of environmental concerns that have significant influence on the overall system behavior but might defer strongly in structural decomposition. In general, three classes can be distinguished: (1) Physical concerns, (2) control concerns and (3) hybrid concerns. Physical concerns only interact directly with the physical system via observation channels. A typical example is the *weather*, which provides sun energy and temperature observations. In contrast, control concerns only interact directly with the control system via information channels. A typical example is the *market*, which provides supply and demand as well as price information. And finally, hybrid concerns interact directly with both the physical system and the control system. A typical example is the *user*, which uses physical and digital interaction.

Detailed modeling the internal structure of environmental concerns is not part of this paper. However, it should be noted that a concise understanding of their interface and behavior is as important for engineering a stable smart grid solution as modeling the control logic itself. Appropriate modeling techniques might vary strongly depending on the domain and the aspects relevant for the overall system behavior.

E. Device Example

In the following all previous aspects are combined into one integrated model of a refrigerator as one example of potential future smart devices. For a typical refrigerator the physical system provides temperature measurements to the control system and the control system triggers cooler activations in the physical system. For our model relevant environmental concerns include the user-door interaction as well as the environmental temperature. Therefore, a user component is connected to the physical system with a door interaction observation channel. In contrast, the environment is connected to the physical system only with a temperature observation channel. Finally, the physical system defines a power output channel.

Semantically, this structural decomposition means the following: (1) The physical system produces power and internal temperature observations depending on cooler activation, door interaction and environment temperature. And (2) the control system produces cooler activations depending on internal temperature measurements. What user and environment observations are dependent on is not further specified.

For the refrigerator the behavior of the physical system is specified by four states, which are derived from the possible combinations of cooler off/on and door closed/open. This state space has been selected since internal temperature measurements vary depending on both cooler activation and door interaction jointly. I.e. cooler activation has potential negative influence on temperature, while door interaction takes influence depending on the environment temperature. In contrast, power observations only vary depending on cooler activation. The transitions between the states are triggered by the control system and the user. It should be noted that for readability the depicted automaton assumes that at a given time point only one of cooler activation and door interaction is allowed to trigger. However, obviously in reality both events are allowed to trigger simultaneously.

Further, the semantics of the automaton is expressed mathematically in Table I. The first two columns contain the valuations of the cooler and door input channels at time point t. The second two columns contain the valuations of the temperature and power output channels instead. If the cooler is `off` the interior temperature is given as a function of the interior temperature at time point $t-1$ and the environment temperature at time point t. If the cooler is `on` the interior temperature is given as a function of the interior temperature

Table I
TABLE SPECIFICATION OF REFRIGERATOR BEHAVIOR.

Cooler(t)	Door(t)	Temperature(t)	Power(t)
`off`	`closed`	$f_1(T(t-1), T_{Env}(t))$	$0W$
`off`	`open`	$f_2(T(t-1), T_{Env}(t))$	$0W$
`on`	`closed`	$f_3(T(t-1))$	$-200W$
`on`	`open`	$f_4(T(t-1))$	$-200W$

at time point $t-1$ neglecting the influence of the environment temperature for simplicity. In both cases the door parameter only influences the rate of temperature change. In contrast, the observed power is only influenced by the cooler input. I.e. the specification of the observed power is analogous to the lamp example, only that the control system is responsible for activation/deactivation rather than the user.

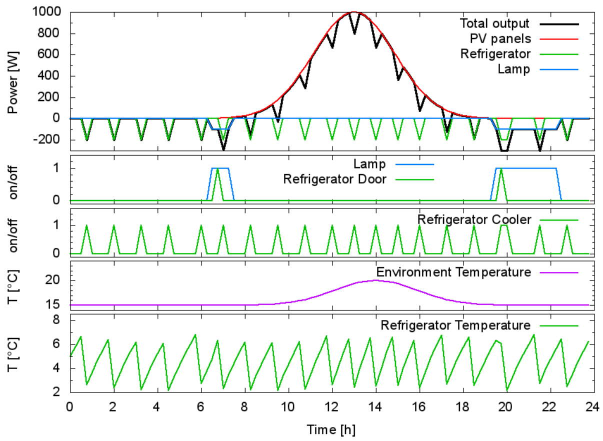
This basic model of the physical system provides the foundation for developing market- and user-driven control algorithms. For example, a simple implementation could (a) neglect the price data, (b) activate the cooler as soon as the measured temperature rises above a certain limit, and (c) deactivate the cooler as soon as the measured temperature falls below a certain limit. However, this strategy might not produce price-optimal control decisions within the user-defined temperature band. Therefore, more advanced algorithms would probably consider the price data during their decision making to prefer activating the cooler when energy is cheap and deactivating the cooler when energy is expensive. When the temperature band is the only defined requirement, both variants produce valid behavioral outcome but might vary strongly in the time of power consumption.

Finally, basic behavioral specifications of the market, the user and the environment can be derived from respective statistics. E.g. typical day profiles could be extracted to prepare a number of scenarios under which the control system can be tested.

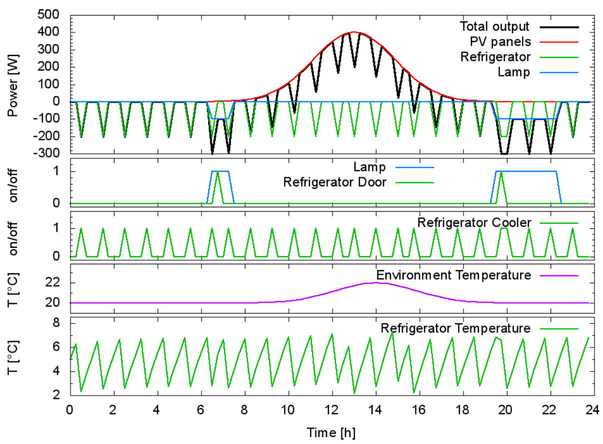
V. SIMULATION RESULTS

Two scenarios have been simulated for evaluating the proposed system model including a lamp, a refrigerator, a solar panel as well as user and environment. The user behavior is identical in both cases and defined by the times when the lamp is switched on and the refrigerator door is opened. The environment behavior is defined by solar power peaking at 13:00 and temperature peaking at 14:00 both being modeled as normal distribution. Scenario 1 assumes high solar power up to $1000W$ and a medium temperature band between $15°C$ and $20°C$. In contrast, Scenario 2 assumes medium solar power up to $400W$ and a high temperature band between $20°C$ and $22°C$. Finally, time is modeled in frames of 15 minutes.

Figure 4 shows the simulated system observations over time. In both scenarios the power output of the lamp is equal due to the constant times of user interaction. Further, the power output of the solar panels only differs by a scale



(a) Scenario 1: High solar power and medium start temperature.



(b) Scenario 2: Medium solar power and high start temperature.

Figure 4. Simulated observations over time for lamp, refrigerator, solar panel, user and environment.

factor due to the different peak values. But the power output of the refrigerator strongly differs from case to case. The reason for this difference is the development of the interior temperature depending on the environment temperature. As in Scenario 2 the environment temperature is higher more cooler activations are needed during the day. A similar effect can be seen in Scenario 1 when comparing day and night time. Overall, the values already resemble typical component behavior even though quite abstract behavioral models are used.

VI. Discussion

The described smart grid model uses a simplified representation of electrical behavior in the physical system. Currently, it considers only active power and neglects physical properties like frequency, voltage, reactive power and network constraints. As demonstrated the simplified model can already be used to describe the control logic, whose goal is to balance the production and consumption of active power. In particular, today's market models are evidence for the model validity as energy is traded in form of kilowatt-hours [16], which are calculated as the integral of active power over time. This analogy further suggests that the method is suited for developing market-related supply and demand balancing mechanisms. However, the simplified model has to be validated in a more extensive study, which is not the scope of this paper.

The simulation demonstrated that the system modeling theory underlying FOCUS is suitable to formalize smart grid systems and gain first results into the behavior of the control logic and its effect on the power balance. These results can be used as a starting point to develop more advanced control mechanisms including optimization of aspects like power balance, energy costs, energy efficiency or consumer priority. Formalizing and analyzing these aspects is a remaining challenge that has not been addressed by this case study.

Further, the current system model only captures a limited class of smart grid components typically found in households. It is expected that the modeling theory is able to represent other device types as well, however this hypothesis has to be validated. Therefore, in the next step the simulation has to be extended to other system components, their respective control logic and environmental concerns.

With other system components also the system complexity has to be increased to analyze the emergent properties of the control logic. Higher system complexity will also mean a more complex interplay between the system components, which can be a major cause of system failures [17]. In principle, FOCUS allows applying formal verification methods to ensure critical system properties across abstraction levels. Providing a respective method that is able to deal with large system scales will be a challenge.

With respect to model analysis one advantage of the suggested physical model is the low computational complexity due to the low number of parameters. However, the pyhsical model only provides a coarse approximation of the underlying phyical state. More precise models typically involve much more parameters and therefore induce worse computational characteristics. To exploit the advantages of both worlds, it has to be investigated which model suffices for which task and how the models can be connected.

Finally, appropriate models for relevant environmental concerns have to be selected. Again, aspects like precision and computational complexity play an important role for the decision. It is expected that statistical models can be used to reduce modeling effort and increase behavioral coverage.

VII. Conclusion & Outlook

Our case study shows that formal software engineering methods have the potential to be applied to the smart grid domain. We have demonstrated that FOCUS is well-suited for formal, multi-disciplinary system modeling. Hereby, in particular the distinction between information and observation channels provides a strong mean for semantically integrating digital and non-digital system aspects. Further, the given example has demonstrated that already a simple model of the electrical system delivers key insights into the balancing behavior of control systems for different scenarios. Finally, the presented simulation results are evidence for the feasibility of the overall approach.

Ongoing research aims at the question how market mechanisms can be used in smart grid scenarios to increase the efficiency of the power system. This includes demand response behavior using different energy prices [18], optimization strategies for load shifting to reduce energy peaks [19], agent-based approaches to model prosumers in the market [20], as well as integration of storage capacity, electrical vehicles and many more. One of our future goals is to integrate these aspects for simulation and formal verification.

Additionally, smart energy systems connecting multiple energy systems such as electrical power, gas or long-distance heat systems gain importance. Their potential to integrate short and long term storages could influence the further development of smart grids. To cover these domains the presented method has to be investigated in terms of development process and provided system models.

Finally, we are aware that current modeling concepts do not suffice to capture all relevant aspects needed for holistic system modeling and thorough model analysis. Therefore, further aspects have to be explored, integrated and validated in order to gain a general understanding of smart grids. Overall, the underlying engineering method has to be tested and adapted continuously to meet all the previously mentioned challenges.

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